

Void Abundance and the Void Size Function with Photometric Redshift Uncertainties:

An Explicit Extension of the Effective-Barrier Excursion-Set Formalism

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Abstract

We provide a self-contained derivation of the excursion-set formalism for void abundance using an effective, scale-dependent (“moving”) barrier, as in recent work (e.g. arXiv:2401.14451). We then extend the formalism to incorporate photometric redshift (photo- z) uncertainties from first principles, starting in Fourier space from the 3D linear matter power spectrum $P(k, z)$. Photometric redshift scatter induces anisotropic *line-of-sight* (LOS) smoothing of the density field; this modifies the smoothing window, the variance $S(R)$, and the cross-covariance Ξ_{ij} that control the correlated random walk. We give explicit, implementation-ready one-dimensional integral expressions for the modified variance $S_{\text{eff}}(R)$, its first and second derivatives with respect to R , and the cross-covariance Ξ_{ij} , all in terms of $P(k)$ and the tophat window. We further motivate and derive a photo- z -aware effective barrier $B_{\text{ph}}(S_{\text{eff}})$ by preserving the normalised peak height $\nu = B/\sqrt{S}$ and discuss additional diffusion D_{ph} . Putting these together yields a modified multiplicity $f_{\text{ph}}(S_{\text{eff}})$ and hence the Eulerian void size function (VSF). Each step is justified and detailed to make the pipeline transparent and directly usable in code.

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1. Preliminaries: notation, conventions, and units

- We work in comoving coordinates. Wavenumbers k are in $h \text{ Mpc}^{-1}$; radii R (Lagrangian) and R_E (Eulerian) are in $h^{-1} \text{ Mpc}$. The linear matter power spectrum is $P(k, z)$ at redshift z .
- Fourier transform convention: the smoothed linear density field at scale R is

$$\delta_R(\mathbf{x}) = \int \frac{d^3 k}{(2\pi)^3} \tilde{\delta}(\mathbf{k}) W_L(kR) e^{i\mathbf{k}\cdot\mathbf{x}}, \quad (1)$$

with a spherically symmetric Lagrangian window $W_L(kR)$; for a real-space tophat of radius R ,

$$W_T(x) \equiv W_L(x) = 3 \frac{\sin x - x \cos x}{x^3}, \quad x \equiv kR. \quad (2)$$

- The isotropic variance and cross-covariance of the smoothed field (correlated steps) are

$$S(R, z) \equiv \sigma^2(R, z) = \int \frac{d^3 k}{(2\pi)^3} P(k, z) |W_L(kR)|^2 = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k, z) |W_L(kR)|^2, \quad (3)$$

$$\Xi(R_i, R_j; z) = \int \frac{d^3 k}{(2\pi)^3} P(k, z) W_L(kR_i) W_L(kR_j) = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k, z) W_L(kR_i) W_L(kR_j). \quad (4)$$

We denote $S_i \equiv \Xi(R_i, R_i) = S(R_i)$.

- Basic excursion-set random walk: as R decreases (filter narrows), $S(R)$ increases; a trajectory $\delta(S)$ performs a correlated Gaussian walk with zero mean and covariance $\langle \delta(S_i) \delta(S_j) \rangle = \Xi(R_i, R_j)$.

2. Effective moving/diffusive barrier for voids

2.1. Motivation

For voids, the linear-theory *spherical* threshold is $\delta_v \simeq -2.71$; however, formation physics, void-finder selection, and non-linear mapping imply that the relevant threshold is *effective* and *scale-dependent*. This motivates a *moving barrier* $B(S)$, often with *diffusion* (stochasticity) to capture additional randomness beyond the Gaussian field.

2.2. Parameterisations and derivatives

A general, flexible form is (paper-specific choices vary; we list two common ones):

$$\text{Power-law drift:} \quad B(S) = B_0 + \beta S^\gamma, \quad (5)$$

$$\text{SMT-like (used in code):} \quad B(S) = \alpha \left[1 + \left(\frac{\beta}{\sqrt{S}} \right)^\gamma \right]. \quad (6)$$

You will need the first few derivatives $B^{(n)}(S) \equiv d^n B/dS^n$ for the multiplicity (see below). For (6), define $u(S) \equiv (\beta/\sqrt{S})^\gamma = \beta^\gamma S^{-\gamma/2}$; then

$$B(S) = \alpha(1 + u), \quad (7)$$

$$B'(S) = \alpha u'(S) = \alpha \beta^\gamma \left(-\frac{\gamma}{2} \right) S^{-\frac{\gamma}{2}-1}, \quad (8)$$

$$B''(S) = \alpha \beta^\gamma \left(-\frac{\gamma}{2} \right) \left(-\frac{\gamma}{2} - 1 \right) S^{-\frac{\gamma}{2}-2}, \quad (9)$$

$$B^{(n)}(S) = \alpha \beta^\gamma \left(-\frac{\gamma}{2} \right) \left(-\frac{\gamma}{2} - 1 \right) \cdots \left(-\frac{\gamma}{2} - (n-1) \right) S^{-\frac{\gamma}{2}-n}. \quad (10)$$

Barrier diffusion is encoded by

$$\text{Var}[B|S] = D_B S, \quad (11)$$

with $D_B \geq 0$ a constant (or weakly scale-dependent) diffusion coefficient.

2.3. First-crossing multiplicity (MB form used for context)

Let $\mathcal{F}(S) dS$ be the probability that a walk first crosses $B(S)$ in $[S, S + dS]$. The multiplicity $f(S)$ (fraction per unit $\ln S$) is

$$f(S) \equiv 2S \mathcal{F}(S). \quad (12)$$

In the moving/diffusive barrier (MB) approximation (and closely related upcrossing forms), a useful, implementation-friendly expression is

$$f(S) \simeq \frac{|T(S)|}{\sqrt{2\pi} S^3} \exp \left[-\frac{B^2(S)}{2(1 + D_B) S} \right], \quad T(S) = \sum_{n=0}^N \frac{(-S)^n}{n!} \frac{d^n B}{dS^n}, \quad (13)$$

with $N \simeq 5-6$ typically sufficient. The exponential term generalises the Markovian $\exp(-\nu^2/2)$ with $\nu \equiv B/\sqrt{S}$ to a diffusing barrier, while the prefactor $T(S)$ encodes barrier drift/curvature and a part of non-Markovian corrections.

2.4. From multiplicity to the void size function

For voids one usually counts per Lagrangian volume element and then maps to Eulerian radii. A compact and widely-used expression is

$$\frac{dn_E}{dR_E} = \frac{3}{4\pi R_E^3} f(S) \left| \frac{dS}{dR} \right| \frac{dR}{dR_E}, \quad (14)$$

where R is Lagrangian and R_E Eulerian, related by a mapping $R_E = q R$ (with q often treated as a constant expansion factor). The Jacobian $3/(4\pi R_E^3)$ arises because one counts voids per Eulerian volume bin.

Derivation sketch of (14). From $f(S)$ as fraction per $d \ln S$, the number density in Lagrangian units scales as $dn_L \propto f(S) d \ln S = f(S) (dS/S)$. Changing variables $S \rightarrow R$ gives $dn_L \propto f(S) |dS/dR| dR/S$. Up to normalisation conventions (volume vs mass weighting for voids), one then maps $R \rightarrow R_E$ and divides by the Eulerian volume element $dV_E = 4\pi R_E^2 dR_E$, yielding (14). In practice, the normalisation is absorbed into f and the chosen mapping.

3. Photometric redshift (photo- z) uncertainties: physics and modelling

3.1. From redshift scatter to LOS smoothing

Photometric redshift errors are commonly modelled as a Gaussian scatter

$$\sigma_z(1+z), \quad (15)$$

which, via the comoving distance-redshift relation, translates to a LOS displacement with rms

$$\sigma_\chi(z) = \left| \frac{d\chi}{dz} \right| \sigma_z(1+z) = \frac{c}{H(z)} \sigma_z(1+z). \quad (16)$$

Thus the reconstructed 3D density field is the true field convolved along the LOS by a Gaussian of width σ_χ . In Fourier space the convolution becomes a product with a Gaussian damping in the $k_\parallel$ direction:

$$\exp\left(-\frac{1}{2} k_\parallel^2 \sigma_\chi^2\right). \quad (17)$$

3.2. Effective anisotropic smoothing window

If the intrinsic filter is $W_L(kR)$ (usually isotropic), the *effective* window with photo- z uncertainty is

$$W_{\text{eff}}(\mathbf{k}; R, z) = W_L(kR) \exp\left(-\frac{1}{2} k_\parallel^2 \sigma_\chi^2(z)\right). \quad (18)$$

This factorisation assumes that the LOS smearing is independent of the angular filter, which is a good approximation when the angular selection is slowly varying along the LOS.

4. Modified variance and covariance: explicit derivations

4.1. Angular integrals and the $G(a)$ factor

We write $\mathbf{k} = (\mathbf{k}_\perp, k_\parallel)$ and use spherical coordinates with $\mu \equiv \cos \theta = k_\parallel/k$. The variance with the anisotropic window is

$$\begin{aligned} S_{\text{eff}}(R, z) &= \int \frac{d^3k}{(2\pi)^3} P(k, z) |W_L(kR)|^2 \exp(-k_\parallel^2 \sigma_\chi^2) \\ &= \frac{1}{(2\pi)^3} \int_0^\infty dk k^2 P(k, z) |W_L(kR)|^2 \int_{-1}^1 d\mu \int_0^{2\pi} d\phi \exp(-k^2 \mu^2 \sigma_\chi^2) \\ &= \frac{1}{(2\pi)^3} \int_0^\infty dk k^2 P(k, z) |W_L(kR)|^2 (2\pi) \underbrace{\int_{-1}^1 d\mu e^{-k^2 \mu^2 \sigma_\chi^2}}_{\mu\text{-integral}}. \end{aligned} \quad (19)$$

The μ -integral is elementary:

$$\int_{-1}^1 d\mu e^{-k^2 \mu^2 \sigma_\chi^2} = \frac{1}{k\sigma_\chi} \int_{-k\sigma_\chi}^{k\sigma_\chi} dt e^{-t^2} = \frac{\sqrt{\pi}}{k\sigma_\chi} \text{erf}(k\sigma_\chi). \quad (20)$$

Therefore

$$\begin{aligned} S_{\text{eff}}(R, z) &= \frac{1}{4\pi^2} \int_0^\infty dk k^2 P(k, z) |W_L(kR)|^2 \left[\frac{\sqrt{\pi}}{k\sigma_\chi} \text{erf}(k\sigma_\chi) \right] \\ &= \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k, z) |W_L(kR)|^2 G(k\sigma_\chi), \end{aligned} \quad (21)$$

where for convenience we defined the *angular damping factor*

$$G(a) \equiv \frac{\sqrt{\pi}}{2a} \text{erf}(a), \quad a \equiv k\sigma_\chi. \quad (22)$$

Note the important limits:

$$a \rightarrow 0 : \quad \text{erf}(a) \sim \frac{2}{\sqrt{\pi}} a \Rightarrow G(a) \rightarrow 1 \Rightarrow S_{\text{eff}} \rightarrow S \text{ (isotropic)}, \quad (23)$$

$$a \rightarrow \infty : \quad \text{erf}(a) \rightarrow 1 \Rightarrow G(a) \sim \frac{\sqrt{\pi}}{2a} \rightarrow 0 \Rightarrow \text{strong LOS damping suppresses high-}k \text{ modes.} \quad (24)$$

Small- σ_χ expansion.. For intuition and quick checks, expand $\text{erf}(a)$: $\text{erf}(a) = \frac{2}{\sqrt{\pi}} \left(a - \frac{a^3}{3} + \frac{a^5}{10} - \dots \right)$.

Hence

$$G(a) = 1 - \frac{a^2}{3} + \frac{a^4}{10} - \dots, \quad (25)$$

and therefore

$$S_{\text{eff}}(R) \simeq S(R) - \frac{\sigma_\chi^2}{3} \int \frac{d^3k}{(2\pi)^3} P |W|^2 k^2 + \frac{\sigma_\chi^4}{10} \int \frac{d^3k}{(2\pi)^3} P |W|^2 k^4 + \dots \quad (26)$$

The leading correction is negative and scales as $\mathcal{O}(\sigma_\chi^2)$.

4.2. Derivatives of $S_{\text{eff}}(R)$ w.r.t. R

Let $x \equiv kR$ and $W_T(x)$ be the tophat in (2). Exact x -derivatives are

$$W'_T(x) = \frac{dW_T}{dx} = 3 \left(\frac{\sin x}{x^2} - \frac{3(\sin x - x \cos x)}{x^4} \right), \quad (27)$$

$$W''_T(x) = \frac{d^2W_T}{dx^2} = 3 \left(\frac{2 \cos x}{x^2} - \frac{5 \sin x}{x^3} + \frac{12(\sin x - x \cos x)}{x^5} \right). \quad (28)$$

Using $d/dR = k d/dx$ and the identity $d(W_T^2)/dR = 2W_T dW_T/dR$, we have

$$\begin{aligned} \frac{dS_{\text{eff}}}{dR} &= \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) \frac{d}{dR} [W_T^2(kR)] G(k\sigma_\chi) \\ &= \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) [2W_T(kR) k W'_T(kR)] G(k\sigma_\chi) \\ &= \frac{1}{\pi^2} \int_0^\infty dk k^3 P(k) W_T(kR) W'_T(kR) G(k\sigma_\chi). \end{aligned} \quad (29)$$

Similarly, using $d^2(W_T^2)/dR^2 = 2[(dW_T/dR)^2 + W_T(d^2W_T/dR^2)]$,

$$\frac{d^2S_{\text{eff}}}{dR^2} = \frac{1}{\pi^2} \int_0^\infty dk k^2 P(k) \left\{ [k W'_T(kR)]^2 + W_T(kR) [k^2 W''_T(kR)] \right\} G(k\sigma_\chi). \quad (30)$$

A dimensionless shape parameter useful in non-Markovian approximations is

$$DW(R) \equiv \frac{\frac{d^2S_{\text{eff}}}{dR^2}}{\left(\frac{dS_{\text{eff}}}{dR}\right)^2}, \quad (31)$$

which reduces to its isotropic counterpart when $\sigma_\chi \rightarrow 0$.

4.3. Cross-covariance Ξ_{ij} with photo- z

For a set of radii $\{R_i\}$, the covariance of the anisotropically filtered field is

$$\begin{aligned} \Xi_{ij}(z) &= \int \frac{d^3k}{(2\pi)^3} P(k, z) W_L(kR_i) W_L(kR_j) e^{-k_\parallel^2 \sigma_\chi^2} \\ &= \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k, z) W_L(kR_i) W_L(kR_j) G(k\sigma_\chi), \end{aligned} \quad (32)$$

with $S_{\text{eff}}(R_i) = \Xi_{ii}$. This Ξ fully specifies the non-Markovian Gaussian process for Monte Carlo walks or for computing slope statistics in upcrossing approximations.

Useful slope covariances for upcrossing.. Let $C(S, S') \equiv \langle \delta(S) \delta(S') \rangle$ be the covariance in S -time. Define the S -derivative $\dot{\delta} \equiv d\delta/dS$. Then

$$\langle \delta \dot{\delta} \rangle = \left. \frac{\partial C(S, S')}{\partial S'} \right|_{S'=S}, \quad \langle \dot{\delta}^2 \rangle = \left. \frac{\partial^2 C(S, S')}{\partial S \partial S'} \right|_{S'=S}. \quad (33)$$

In practice, on a grid $\{R_i\}$ (hence $\{S_i\}$), one estimates these by finite differences from Ξ_{ij} after mapping $R \rightarrow S_{\text{eff}}(R)$.

5. A photo- z adjusted effective barrier

5.1. Preserving the normalised peak height

Photo- z damping reduces the two-point variance at fixed R : $S_{\text{eff}}(R) < S(R)$. If the physical/selection threshold is encoded in a barrier $B(S)$ fitted in (near-)spectroscopic conditions, a natural parameter-free mapping that preserves the *significance* (peak height)

$$\nu \equiv \frac{B}{\sqrt{S}}, \quad (34)$$

is to demand $\nu_{\text{ph}} = \nu$, i.e.

$$\frac{B_{\text{ph}}(R)}{\sqrt{S_{\text{eff}}(R)}} = \frac{B(S(R))}{\sqrt{S(R)}} \Rightarrow \boxed{B_{\text{ph}}(R) = \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} B(S(R))}. \quad (35)$$

By construction, as $\sigma_\chi \rightarrow 0$ one has $S_{\text{eff}} \rightarrow S$ and $B_{\text{ph}} \rightarrow B$.

5.2. Optional nuisance shifts and added diffusion

Finder/selection changes between spectroscopic-like and photometric catalogues may induce small systematic shifts. Allow

$$\boxed{B_{\text{ph}}(R) = \alpha_B \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} B(S(R)) + \beta_B}, \quad (36)$$

with $(\alpha_B, \beta_B) = (1, 0)$ for a pure theory prediction, or treated as weakly informative nuisances calibrated on photometric mocks. Likewise, variability in photo- z kernels across objects adds effective diffusion,

$$\boxed{D_{\text{tot}} = D_B + D_{\text{ph}}}, \quad D_{\text{ph}} \geq 0, \quad (37)$$

with $D_{\text{ph}} = 0$ in the simplest case of a homogeneous Gaussian kernel.

5.3. Derivatives $d^n B_{\text{ph}}/dS_{\text{eff}}^n$ by chain rule

Most multiplicity formulae (MB/upcrossing) require B_{ph} and its derivatives with respect to S_{eff} . It is numerically stable to regard all quantities as functions of R and use

$$\frac{dB_{\text{ph}}}{dS_{\text{eff}}} = \frac{dB_{\text{ph}}/dR}{dS_{\text{eff}}/dR}, \quad \frac{d^2 B_{\text{ph}}}{dS_{\text{eff}}^2} = \frac{1}{(dS_{\text{eff}}/dR)^2} \left(\frac{d^2 B_{\text{ph}}}{dR^2} - \frac{dB_{\text{ph}}}{dR} \frac{d^2 S_{\text{eff}}}{dR^2} / \frac{dS_{\text{eff}}}{dR} \right), \quad (38)$$

and similarly for higher derivatives. From (35), the R -derivative of the deterministic prefactor is

$$\frac{d}{dR} \sqrt{\frac{S_{\text{eff}}}{S}} = \frac{1}{2} \sqrt{\frac{S}{S_{\text{eff}}}} \left(\frac{1}{S_{\text{eff}}} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S} \frac{dS}{dR} \right), \quad (39)$$

where dS/dR is obtained by setting $G \equiv 1$ in (29)–(30).

Derivative of the square-root ratio $\sqrt{S_{\text{eff}}/S}$ with respect to R

We aim to derive

$$\frac{d}{dR} \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} = \frac{1}{2} \sqrt{\frac{S(R)}{S_{\text{eff}}(R)}} \left(\frac{1}{S_{\text{eff}}(R)} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S(R)} \frac{dS}{dR} \right). \quad (40)$$

For brevity, we write $S_{\text{eff}} \equiv S_{\text{eff}}(R)$ and $S \equiv S(R)$ when no confusion arises.

Step 1: Define the function and use the chain rule.. Let

$$F(R) \equiv \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} = \left(\frac{S_{\text{eff}}(R)}{S(R)} \right)^{1/2}. \quad (41)$$

Using the chain rule for a composite function $u(R)^{1/2}$ with $u(R) \equiv S_{\text{eff}}(R)/S(R)$:

$$\frac{dF}{dR} = \frac{1}{2} u(R)^{-1/2} \frac{du}{dR} = \frac{1}{2} \left(\frac{S_{\text{eff}}}{S} \right)^{-1/2} \frac{d}{dR} \left(\frac{S_{\text{eff}}}{S} \right). \quad (42)$$

Step 2: Differentiate the ratio $u(R) = S_{\text{eff}}(R)/S(R)$.. Compute the derivative of a ratio using the quotient rule:

$$\frac{d}{dR} \left(\frac{S_{\text{eff}}}{S} \right) = \frac{S \frac{dS_{\text{eff}}}{dR} - S_{\text{eff}} \frac{dS}{dR}}{S^2}. \quad (43)$$

Therefore,

$$\frac{dF}{dR} = \frac{1}{2} \left(\frac{S_{\text{eff}}}{S} \right)^{-1/2} \frac{S \frac{dS_{\text{eff}}}{dR} - S_{\text{eff}} \frac{dS}{dR}}{S^2}. \quad (44)$$

Step 3: Simplify the prefactor $\left(\frac{S_{\text{eff}}}{S} \right)^{-1/2}$.. We have

$$\left(\frac{S_{\text{eff}}}{S} \right)^{-1/2} = \sqrt{\frac{S}{S_{\text{eff}}}}. \quad (45)$$

Substitute and rearrange:

$$\frac{dF}{dR} = \frac{1}{2} \sqrt{\frac{S}{S_{\text{eff}}}} \frac{S \frac{dS_{\text{eff}}}{dR} - S_{\text{eff}} \frac{dS}{dR}}{S^2} \quad (46)$$

$$= \frac{1}{2} \sqrt{\frac{S}{S_{\text{eff}}}} \left(\frac{1}{S} \frac{dS_{\text{eff}}}{dR} - \frac{S_{\text{eff}}}{S^2} \frac{dS}{dR} \right). \quad (47)$$

Step 4: Express in terms of $\frac{1}{S_{\text{eff}}}$ and $\frac{1}{S}$. Note that

$$\frac{S_{\text{eff}}}{S^2} = \left(\frac{S_{\text{eff}}}{S} \right) \frac{1}{S} = \left(\sqrt{\frac{S_{\text{eff}}}{S}} \right)^2 \frac{1}{S}. \quad (48)$$

Multiplying by the prefactor $\sqrt{S/S_{\text{eff}}}$ gives

$$\sqrt{\frac{S}{S_{\text{eff}}}} \frac{S_{\text{eff}}}{S^2} = \sqrt{\frac{S_{\text{eff}}}{S}} \frac{1}{S} = \frac{1}{\sqrt{S_{\text{eff}}S}}. \quad (49)$$

Similarly,

$$\sqrt{\frac{S}{S_{\text{eff}}}} \frac{1}{S} = \frac{1}{\sqrt{S_{\text{eff}}S}} \frac{1}{\sqrt{S}}. \quad (50)$$

Using these, we can write

$$\frac{dF}{dR} = \frac{1}{2} \left[\sqrt{\frac{S}{S_{\text{eff}}}} \frac{1}{S} \frac{dS_{\text{eff}}}{dR} - \sqrt{\frac{S}{S_{\text{eff}}}} \frac{S_{\text{eff}}}{S^2} \frac{dS}{dR} \right] \quad (51)$$

$$= \frac{1}{2} \left[\sqrt{\frac{S}{S_{\text{eff}}}} \frac{1}{S} \frac{dS_{\text{eff}}}{dR} - \frac{1}{\sqrt{S_{\text{eff}}S}} \frac{dS}{dR} \right]. \quad (52)$$

Factorising $\sqrt{S/S_{\text{eff}}}$ and rewriting in a symmetric form yields the compact expression:

$$\boxed{\frac{d}{dR} \sqrt{\frac{S_{\text{eff}}}{S}} = \frac{1}{2} \sqrt{\frac{S}{S_{\text{eff}}}} \left(\frac{1}{S_{\text{eff}}} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S} \frac{dS}{dR} \right)}. \quad (53)$$

Alternative derivation by logarithmic differentiation (optional).. Let $F = \sqrt{S_{\text{eff}}/S}$, then $\ln F = \frac{1}{2}(\ln S_{\text{eff}} - \ln S)$. Differentiate:

$$\frac{1}{F} \frac{dF}{dR} = \frac{1}{2} \left(\frac{1}{S_{\text{eff}}} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S} \frac{dS}{dR} \right), \quad (54)$$

hence

$$\frac{dF}{dR} = F \frac{1}{2} \left(\frac{1}{S_{\text{eff}}} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S} \frac{dS}{dR} \right) = \frac{1}{2} \sqrt{\frac{S_{\text{eff}}}{S}} \left(\frac{1}{S_{\text{eff}}} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S} \frac{dS}{dR} \right). \quad (55)$$

Multiplying and dividing by $\sqrt{S/S_{\text{eff}}}$ gives the same boxed result above.

6. Peak height $\nu \equiv B/\sqrt{S}$: definition and motivation

Gaussian smoothed density and variance.. Let δ_R be the linearly evolved, smoothed density contrast at Lagrangian scale R . For a fixed R , δ_R is a Gaussian random variable with mean zero and variance

$$S(R) \equiv \sigma^2(R) = \int \frac{d^3k}{(2\pi)^3} P(k) |W_L(kR)|^2, \quad (56)$$

where $P(k)$ is the linear matter power spectrum and W_L is the Fourier-space smoothing window (e.g. tophat).

Barrier crossing and rarity.. In excursion-set theory, structure formation or void selection is encoded by a *barrier* $B(S)$: a trajectory $\delta(S)$ forms the object/void when it *first crosses* $\delta = B(S)$. For a Gaussian field, the probability density of δ_R at variance S is

$$p(\delta_R; S) = \frac{1}{\sqrt{2\pi S}} \exp\left(-\frac{\delta_R^2}{2S}\right). \quad (57)$$

The rarity of a fluctuation equal to the barrier is governed by the combination $B^2/(2S)$ in the exponential.

Definition of peak height.. The *peak height* ν is the dimensionless ratio

$$\boxed{\nu(S) \equiv \frac{B(S)}{\sqrt{S}}}, \quad (58)$$

so that the Gaussian suppression reads $\exp(-\nu^2/2)$. This rescales out the variance S , making ν the natural measure of how “rare” or “significant” the threshold is relative to the typical fluctuations at that scale.

Why ν is the right variable..

- *Scale invariance:* ν compares the threshold to the rms fluctuation, allowing meaningful comparison across different smoothing scales R (or variances S).
- *Universality:* Many excursion-set predictions and empirical mass/void functions take approximately universal forms when expressed in terms of ν , because the leading Gaussian factor depends only on ν^2 .
- *Moving barriers:* Even when B depends on S , the leading suppression remains $\exp[-B^2/(2S)] = \exp(-\nu^2/2)$, while corrections enter through barrier derivatives $B'(S), B''(S)$ in the multiplicity prefactor.

Halos vs. voids.. For halos, $B \approx \delta_c > 0$, so $\nu = \delta_c/\sqrt{S}$ grows as S decreases (rare massive halos). For voids, B is typically negative (e.g. $B \approx \delta_v < 0$ or an effective barrier); the sign affects the crossing direction, but the Gaussian suppression depends on ν^2 , hence on $|B|/\sqrt{S}$.

Photo- z extension (variance change).. Photo- z LOS smoothing reduces the variance at fixed R (i.e. $S_{\text{eff}} < S$). To preserve threshold significance, one may demand

$$\nu_{\text{ph}} \equiv \frac{B_{\text{ph}}}{\sqrt{S_{\text{eff}}}} \stackrel{!}{=} \nu \equiv \frac{B}{\sqrt{S}} \quad \Rightarrow \quad B_{\text{ph}}(R) = \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} B(S(R)), \quad (59)$$

which keeps the Gaussian factor $\exp(-\nu^2/2)$ unchanged while consistently accounting for the modified variance due to photo- z effects.

7. Barrier diffusion parameters: D_{tot} , D_B , and D_{ph}

In excursion-set treatments with an *effective* (moving) barrier, the crossing condition is not perfectly deterministic. Various physical and observational effects introduce *stochasticity* in the barrier value at a given variance S , which broadens the first-crossing statistics. This stochasticity is conveniently captured by *barrier diffusion* parameters.

Baseline barrier diffusion D_B . Even in spectroscopic-quality settings (i.e. without photometric redshift uncertainties), void formation/selection is subject to randomness that is not fully represented by a single deterministic threshold. Examples include:

- departures from spherical symmetry and environment-dependent dynamics;
- finder-induced selection effects (centering uncertainty, merging/splitting of voids);
- residual stochasticity in mapping from the linear theory threshold (e.g. δ_v) to the effective barrier used by the void finder.

These effects are encoded by a *baseline* diffusion coefficient $D_B \geq 0$ through

$$\text{Var}[B | S] = D_B S, \quad (60)$$

meaning that, at fixed “time” S , the barrier value fluctuates with a variance proportional to S . In multiplicity formulae (e.g. moving/diffusive barrier or upcrossing approximations), D_B *reduces* the effective Gaussian suppression, typically entering the exponential as

$$\exp\left[-\frac{B^2(S)}{2S}\right] \longrightarrow \exp\left[-\frac{B^2(S)}{2(1+D_B)S}\right], \quad (61)$$

which increases the probability of barrier crossings (larger $f(S)$), especially at small S where the deterministic barrier would be very “rare”.

Photo- z -induced diffusion D_{ph} . Photometric redshift uncertainties add *additional* sources of stochasticity specific to the reconstruction of the density field along the line of sight:

- object-to-object variability in photo- z kernel widths or shapes (per-galaxy PDFs);
- non-Gaussian tails, outliers, and catastrophic failures (extreme redshift misestimates);
- spatial or selection dependence in redshift performance that makes the effective threshold fuzzier.

These effects cannot be represented solely by a *deterministic* change in the variance $S \rightarrow S_{\text{eff}}$; they introduce *extra randomness*. We therefore add a photo- z diffusion coefficient $D_{\text{ph}} \geq 0$, again as

$$\text{Var}[B | S_{\text{eff}}] = D_{\text{ph}} S_{\text{eff}}. \quad (62)$$

In practice, D_{ph} behaves like a *nuisance parameter* calibrated on photometric mocks, because it depends on survey-specific photo- z performance and the details of the void finder.

Total diffusion D_{tot} . The net barrier stochasticity in photometric settings is the *sum* of the baseline and photo- z contributions:

$$\boxed{D_{\text{tot}} = D_B + D_{\text{ph}}}. \quad (63)$$

One then uses D_{tot} in the multiplicity formulae. For example, in the upcrossing form with a photo- z adjusted barrier $B_{\text{ph}}(S_{\text{eff}})$, the leading exponential suppression becomes

$$\exp\left[-\frac{B_{\text{ph}}^2(S_{\text{eff}})}{2S_{\text{eff}}}\right] \longrightarrow \exp\left[-\frac{B_{\text{ph}}^2(S_{\text{eff}})}{2(1+D_{\text{tot}})S_{\text{eff}}}\right], \quad (64)$$

and any slope-variance terms (e.g. $\Gamma_{\delta\delta}$ in the upcrossing prefactor) are computed consistently from the photo- z covariance. Conceptually, D_{tot} controls how “fuzzy” the crossing condition is: larger D_{tot} broadens the distribution of barrier values and increases the crossing rate, thereby moderating the steepness of the multiplicity at small S .

Practical guidance..

- *Theory-only predictions:* set $D_{\text{ph}} = 0$ and use D_B calibrated on spectroscopic mocks (or a small value consistent with the effective-barrier literature).
- *Photometric analyses:* treat D_{ph} as a weakly informative nuisance, constrained by photometric mocks and/or cross-validation on data; propagate $D_{\text{tot}} = D_B + D_{\text{ph}}$ through the chosen multiplicity formula.
- *Consistency limit:* when photo- z scatter vanishes ($\sigma_\chi \rightarrow 0$), $S_{\text{eff}} \rightarrow S$, $D_{\text{ph}} \rightarrow 0$, and all expressions reduce to the spectroscopic baseline.

8. Photo- z multiplicity and the VSF

8.1. MB/upcrossing family with photo- z (contextual form)

Replace $S \rightarrow S_{\text{eff}}$ and $B \rightarrow B_{\text{ph}}$ in (13), and use D_{tot} :

$$f_{\text{ph}}(S_{\text{eff}}) \simeq \frac{|T_{\text{ph}}(S_{\text{eff}})|}{\sqrt{2\pi S_{\text{eff}}^3}} \exp\left[-\frac{B_{\text{ph}}^2(S_{\text{eff}})}{2(1 + D_{\text{tot}}) S_{\text{eff}}}\right], \quad (65)$$

$$T_{\text{ph}}(S_{\text{eff}}) = \sum_{n=0}^N \frac{(-S_{\text{eff}})^n}{n!} \frac{d^n B_{\text{ph}}}{dS_{\text{eff}}^n}. \quad (66)$$

If your implementation uses an alternative (e.g. a specific non-Markovian normalisation factor involving DW), compute that factor with S_{eff} and derivatives (29)–(30).

8.2. Eulerian void size function

The final expression mirrors (14) with $S \rightarrow S_{\text{eff}}$:

$$\frac{dn_{\text{E}}}{dR_{\text{E}}} = \frac{3}{4\pi R_{\text{E}}^3} f_{\text{ph}}(S_{\text{eff}}) \left| \frac{dS_{\text{eff}}}{dR} \right| \left| \frac{dR}{dR_{\text{E}}} \right|. \quad (67)$$

Adopt your mapping $R_{\text{E}}(R)$ (e.g. $R_{\text{E}} = qR$) and corresponding Jacobian dR/dR_{E} .

9. First-crossing multiplicity: upcrossing value–slope Gaussian approximation

We use the upcrossing approximation for the first-crossing multiplicity $f(S)$, written in terms of the barrier value $B_S \equiv B(S)$, its derivative $B'_S \equiv dB/dS$, and the value–slope covariance combination $\Gamma_{\delta\delta}$ derived from the correlated Gaussian walk. The multiplicity is

$$f(S) \simeq \frac{e^{-B_S^2/(2S)}}{\sqrt{2\pi S}} \left[\sqrt{\frac{\Gamma_{\delta\delta}}{2\pi S}} \exp\left(-\frac{S}{2\Gamma_{\delta\delta}} \left(\frac{B_S}{2S} - B'_S\right)^2\right) + \frac{1}{2} \left(\frac{B_S}{2S} - B'_S\right) \left\{ \text{erf}\left[\sqrt{\frac{S}{2\Gamma_{\delta\delta}}} \left(\frac{B_S}{2S} - B'_S\right)\right] + 1 \right\} \right]. \quad (68)$$

This expression follows from the joint Gaussian statistics of the walk value δ and its “slope” $\dot{\delta} \equiv d\delta/dS$, evaluated at the moving barrier with an absorbing boundary and counting upcrossings. The quantity $\Gamma_{\delta\delta}$ is built from the covariance kernel $C(S, S') = \langle \delta(S)\delta(S') \rangle$ (see below).

Barrier functional form.. We adopt

$$B(S) = \alpha \left[1 + \left(\frac{\beta}{S} \right)^\gamma \right], \quad B'(S) = \alpha \beta^\gamma (-\gamma) S^{-(\gamma+1)}. \quad (69)$$

Photo- z extension.. Photometric redshift (photo- z) uncertainties modify the smoothing along the line of sight (LOS), yielding a *photo- z effective* variance $S_{\text{eff}}(R)$ and covariance Ξ^{eff} through the LOS damping factor $G(k\sigma_\chi)$. To consistently include photo- z in the upcrossing multiplicity, replace:

$$S \rightarrow S_{\text{eff}}, \quad B \rightarrow B_{\text{ph}}, \quad B' \rightarrow \frac{dB_{\text{ph}}}{dS_{\text{eff}}}, \quad \Gamma_{\delta\delta} \rightarrow \Gamma_{\delta\delta}^{\text{eff}}. \quad (70)$$

The photo- z adjusted barrier preserves the peak height $\nu \equiv B/\sqrt{S}$ and allows for small nuisance shifts:

$$B_{\text{ph}}(R) = \alpha_B \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} B(S(R)) + \beta_B, \quad (71)$$

with $(\alpha_B, \beta_B) = (1, 0)$ for a parameter-free theory prediction. Its derivative with respect to S_{eff} is obtained via the chain rule through R :

$$\frac{dB_{\text{ph}}}{dS_{\text{eff}}} = \frac{\frac{dB_{\text{ph}}}{dR}}{\frac{dS_{\text{eff}}}{dR}}, \quad \frac{dB_{\text{ph}}}{dR} = \alpha_B \left[\frac{d}{dR} \sqrt{\frac{S_{\text{eff}}}{S}} B(S) + \sqrt{\frac{S_{\text{eff}}}{S}} \frac{dB}{dS} \frac{dS}{dR} \right], \quad (72)$$

where

$$\frac{d}{dR} \sqrt{\frac{S_{\text{eff}}}{S}} = \frac{1}{2} \sqrt{\frac{S}{S_{\text{eff}}}} \left(\frac{1}{S_{\text{eff}}} \frac{dS_{\text{eff}}}{dR} - \frac{1}{S} \frac{dS}{dR} \right), \quad (73)$$

and dS/dR is the isotropic derivative (i.e. with $G=1$).

Photo- z upcrossing formula.. With these replacements, the photo- z multiplicity becomes

$$f_{\text{ph}}(S_{\text{eff}}) \simeq \frac{e^{-B_{\text{ph}}^2/(2S_{\text{eff}})}}{\sqrt{2\pi S_{\text{eff}}}} \left[\sqrt{\frac{\Gamma_{\delta\delta}^{\text{eff}}}{2\pi S_{\text{eff}}}} \exp\left(-\frac{S_{\text{eff}}}{2\Gamma_{\delta\delta}^{\text{eff}}} \left(\frac{B_{\text{ph}}}{2S_{\text{eff}}} - \frac{dB_{\text{ph}}}{dS_{\text{eff}}} \right)^2 \right) + \frac{1}{2} \left(\frac{B_{\text{ph}}}{2S_{\text{eff}}} - \frac{dB_{\text{ph}}}{dS_{\text{eff}}} \right) \left\{ \text{erf} \left[\sqrt{\frac{S_{\text{eff}}}{2\Gamma_{\delta\delta}^{\text{eff}}}} \left(\frac{B_{\text{ph}}}{2S_{\text{eff}}} - \frac{dB_{\text{ph}}}{dS_{\text{eff}}} \right) \right] + 1 \right\} \right]. \quad (74)$$

Construction of $\Gamma_{\delta\delta}^{\text{eff}}$.. Let $C(S, S') = \langle \delta(S) \delta(S') \rangle$ be the “time” covariance of the correlated Gaussian walk, built from the photo- z covariance kernel $\Xi^{\text{eff}}(R_i, R_j)$ via the mapping $R \leftrightarrow S_{\text{eff}}(R)$:

$$\Xi_{ij}^{\text{eff}} = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) W_T(kR_i) W_T(kR_j) G(k\sigma_\chi), \quad S_{\text{eff}}(R_i) = \Xi_{ii}^{\text{eff}}. \quad (75)$$

Then the value-slope moments follow by finite differences in S :

$$\langle \delta^2 \rangle = S, \quad (76)$$

$$\langle \delta \dot{\delta} \rangle = \frac{\partial C(S, S')}{\partial S'} \Big|_{S'=S}, \quad \dot{\delta} \equiv \frac{d\delta}{dS}, \quad (77)$$

$$\langle \dot{\delta}^2 \rangle = \frac{\partial^2 C(S, S')}{\partial S \partial S'} \Big|_{S'=S}. \quad (78)$$

The combination $\Gamma_{\delta\delta}^{\text{eff}}$ used in (74) is obtained from these moments (see the original upcrossing derivation). In practice, compute Ξ^{eff} on a grid of radii, map to S_{eff} , and estimate the derivatives by finite differences to form $\Gamma_{\delta\delta}^{\text{eff}}$.

From f_{ph} to the Eulerian VSF.. With the Lagrangian→Eulerian mapping $R_{\text{E}} \approx q R$, the Eulerian void size function is

$$\frac{dn_{\text{E}}}{dR_{\text{E}}} = \frac{3}{4\pi R_{\text{E}}^3} f_{\text{ph}}(S_{\text{eff}}) \left| \frac{dS_{\text{eff}}}{dR} \right| \frac{dR}{dR_{\text{E}}}. \quad (79)$$

This reduces exactly to the spectroscopic case when $\sigma_{\chi} \rightarrow 0$ (hence $G \rightarrow 1$, $S_{\text{eff}} \rightarrow S$, $B_{\text{ph}} \rightarrow B$, and $\Gamma_{\delta\delta}^{\text{eff}} \rightarrow \Gamma_{\delta\delta}$).

10. Generalisation: non-Gaussian or heterogeneous photo- z kernels

If the LOS kernel is not Gaussian or varies across objects, denote its (unit-normalised) LOS profile by $K(\chi)$ and its Fourier transform along the LOS by $\tilde{K}(k_{\parallel})$. The anisotropic window becomes

$$W_{\text{eff}}(\mathbf{k}; R) = W_L(kR) \tilde{K}(k_{\parallel}), \quad (80)$$

and the only change in our formulas is the replacement

$$G(k\sigma_{\chi}) \longrightarrow \mathcal{G}(k) \equiv \frac{1}{2} \int_{-1}^1 d\mu |\tilde{K}(k\mu)|^2, \quad (81)$$

so that

$$S_{\text{eff}}(R) = \frac{1}{2\pi^2} \int_0^{\infty} dk k^2 P(k) |W_L(kR)|^2 \mathcal{G}(k), \quad (82)$$

$$\Xi_{ij} = \frac{1}{2\pi^2} \int_0^{\infty} dk k^2 P(k) W_L(kR_i) W_L(kR_j) \mathcal{G}(k). \quad (83)$$

For a Gaussian kernel with width σ_{χ} , $|\tilde{K}|^2 = e^{-k_{\parallel}^2 \sigma_{\chi}^2}$ and $\mathcal{G}(k) = G(k\sigma_{\chi})$ in (22). If a population has a distribution of widths $p(\sigma_{\chi})$, then

$$\mathcal{G}(k) = \frac{1}{2} \int_{-1}^1 d\mu \int d\sigma_{\chi} p(\sigma_{\chi}) e^{-k^2 \mu^2 \sigma_{\chi}^2} = \int d\sigma_{\chi} p(\sigma_{\chi}) G(k\sigma_{\chi}), \quad (84)$$

which is *not* equal to $G(k\sqrt{\langle\sigma_{\chi}^2\rangle})$ (Jensen), providing a principled route for modelling D_{ph} phenomenology if desired.

11. Implementation notes, checks, and numerical guidance

- **Units:** ensure $P(k)$, k , and R are in consistent h -units (k in $h \text{ Mpc}^{-1}$, R in $h^{-1} \text{ Mpc}$). For σ_{χ} , use $c/H(z)$ in Mpc and multiply by h to get $h^{-1} \text{ Mpc}$ if your k carries h .
- **k -range and sampling:** log-sample $k \in [10^{-4}, 10] h \text{ Mpc}^{-1}$ (wider if needed), interpolate $P(k)$ smoothly, and use the analytic $G(k\sigma_{\chi})$ to avoid 2D $(k_{\perp}, k_{\parallel})$ grids.
- **Derivatives:** prefer analytic $W'_T(x)$, $W''_T(x)$ and (29)–(30) over finite differences; this stabilises DW and the barrier-derivative terms.

- $\sigma_\chi \rightarrow 0$ **check:** verify $G \rightarrow 1$ numerically so that $S_{\text{eff}} \rightarrow S$, $\Xi \rightarrow \Xi_{\text{iso}}$, $B_{\text{ph}} \rightarrow B$, $f_{\text{ph}} \rightarrow f$, and the VSF reduces to the spectroscopic prediction.
- **Monte Carlo walks (optional):** draw $\delta \sim \mathcal{N}(0, \Xi)$ using Cholesky $\Xi = LL^\top$. For *voids*, the barrier is *negative* and one counts *downward* crossings as S increases (i.e. as R decreases). Order the grid from small S (large R) to large S (small R). Regularise Ξ by a tiny diagonal if nearly singular.
- **Barrier calibration:** for forecasts, set $(\alpha_B, \beta_B, D_{\text{ph}}) = (1, 0, 0)$; for data analyses, calibrate on photometric mocks and marginalise with weak priors.

12. Worked summary: equations to implement

Tophat and derivatives (with $x = kR$)..

$$W_T(x) = 3 \frac{\sin x - x \cos x}{x^3}, \quad W'_T(x) = 3 \left(\frac{\sin x}{x^2} - \frac{3(\sin x - x \cos x)}{x^4} \right), \quad (85)$$

$$W''_T(x) = 3 \left(\frac{2 \cos x}{x^2} - \frac{5 \sin x}{x^3} + \frac{12(\sin x - x \cos x)}{x^5} \right). \quad (86)$$

Angular factor..

$$G(a) \equiv \frac{\sqrt{\pi}}{2a} \text{erf}(a), \quad a = k\sigma_\chi, \quad G(0) = 1. \quad (87)$$

Variance and derivatives..

$$S_{\text{eff}}(R) = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) W_T^2(kR) G(k\sigma_\chi), \quad (88)$$

$$\frac{dS_{\text{eff}}}{dR} = \frac{1}{\pi^2} \int_0^\infty dk k^3 P(k) W_T(kR) W'_T(kR) G(k\sigma_\chi), \quad (89)$$

$$\frac{d^2 S_{\text{eff}}}{dR^2} = \frac{1}{\pi^2} \int_0^\infty dk k^2 P(k) \left\{ [kW'_T(kR)]^2 + W_T(kR) [k^2 W''_T(kR)] \right\} G(k\sigma_\chi), \quad (90)$$

$$DW(R) = \frac{d^2 S_{\text{eff}}/dR^2}{(dS_{\text{eff}}/dR)^2}. \quad (91)$$

Cross-covariance..

$$\Xi_{ij} = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) W_T(kR_i) W_T(kR_j) G(k\sigma_\chi). \quad (92)$$

Barrier with photo- z ..

$$B_{\text{ph}}(R) = \alpha_B \sqrt{\frac{S_{\text{eff}}(R)}{S(R)}} B(S(R)) + \beta_B, \quad (93)$$

$$\frac{dB_{\text{ph}}}{dS_{\text{eff}}} = \frac{dB_{\text{ph}}/dR}{dS_{\text{eff}}/dR}, \quad D_{\text{tot}} = D_B + D_{\text{ph}}. \quad (94)$$

Multiplicity and VSF (MB form)..

$$f_{\text{ph}}(S_{\text{eff}}) \simeq \frac{|T_{\text{ph}}(S_{\text{eff}})|}{\sqrt{2\pi S_{\text{eff}}^3}} \exp\left[-\frac{B_{\text{ph}}^2(S_{\text{eff}})}{2(1 + D_{\text{tot}}) S_{\text{eff}}}\right], \quad (95)$$

$$T_{\text{ph}}(S_{\text{eff}}) = \sum_{n=0}^N \frac{(-S_{\text{eff}})^n}{n!} \frac{d^n B_{\text{ph}}}{dS_{\text{eff}}^n}, \quad (N \simeq 5-6), \quad (96)$$

$$\frac{dn_{\text{E}}}{dR_{\text{E}}} = \frac{3}{4\pi R_{\text{E}}^3} f_{\text{ph}}(S_{\text{eff}}) \left| \frac{dS_{\text{eff}}}{dR} \right| \frac{dR}{dR_{\text{E}}}. \quad (97)$$

Multiplicity and VSF (upcrossing form)..

$$f_{\text{ph}}(S_{\text{eff}}) \simeq \frac{e^{-B_{\text{ph}}^2/(2S_{\text{eff}})}}{\sqrt{2\pi S_{\text{eff}}}} \left[\sqrt{\frac{\Gamma_{\delta\delta}^{\text{eff}}}{2\pi S_{\text{eff}}}} \exp\left(-\frac{S_{\text{eff}}}{2\Gamma_{\delta\delta}^{\text{eff}}} \left(\frac{B_{\text{ph}}}{2S_{\text{eff}}} - \frac{dB_{\text{ph}}}{dS_{\text{eff}}}\right)^2\right) + \frac{1}{2} \left(\frac{B_{\text{ph}}}{2S_{\text{eff}}} - \frac{dB_{\text{ph}}}{dS_{\text{eff}}}\right) \left\{ \text{erf}\left[\sqrt{\frac{S_{\text{eff}}}{2\Gamma_{\delta\delta}^{\text{eff}}}} \left(\frac{B_{\text{ph}}}{2S_{\text{eff}}} - \frac{dB_{\text{ph}}}{dS_{\text{eff}}}\right)\right] + 1 \right\} \right], \quad (98)$$

$$\frac{dn_{\text{E}}}{dR_{\text{E}}} = \frac{3}{4\pi R_{\text{E}}^3} f_{\text{ph}}(S_{\text{eff}}) \left| \frac{dS_{\text{eff}}}{dR} \right| \frac{dR}{dR_{\text{E}}}. \quad (99)$$

13. Discussion and caveats

- The *only* first-principles change from photo- z is the LOS damping in (18), which mathematically becomes the angular factor $G(k\sigma_{\chi})$ in all k -integrals. Everything else (barrier machinery, non-Markovian corrections) remains structurally identical after $S \rightarrow S_{\text{eff}}$, $\Xi \rightarrow \Xi_{\text{eff}}$.
- The barrier prescription (35) preserves the peak-height ν and is parameter-free. In real catalogues, finder/systematics induce small shifts; (36) and D_{ph} capture these and can be calibrated on photometric mocks.
- The formalism reduces exactly to the spectroscopic case as $\sigma_{\chi} \rightarrow 0$, providing a stringent internal consistency check.
- For very large σ_{χ} (loss of radial information), a projected-void formalism (2D smoothing on the sphere or cylindrical projection) may be more appropriate.